Prior Covid-19 Infection Is Associated With Performance Impairment in a Speeded-Response Priming Task

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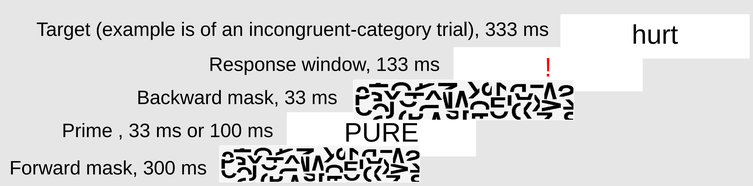
Introduction

- Unconscious category priming occurs when a briefly flashed, visually-masked, consciously imperceptible prime word influences categorization of a visible target word that follows it after a short interval (in the experiments reported here, 33 ms).
- Priming involves semantic associations, which reside mainly in frontal cortex. COVID-19 infection has been found to result in mid- or long-term neurocognitive deficits in this region (Douaud et al., 2022).
- Individuals who have contracted COVID-19, regardless of the severity of their symptoms, can experience these cognitive impairments (Nouraeinejad, 2022). Infection of COVID-19 is correlated with detriments to executive functioning, processing speed, attention, verbal learning and memory, and working memory (Tavares-Júnior et al., 2022).
- Here, we focused on the possibility of observing deficits in priming as sequelae of prior COVID-19 infection. Participants with prior infections were expected to show less priming. Our research also explored the possibility of more global deficits in performance in the priming task.

Method

- Participants first gained practice categorizing targets not preceded by primes. They then did a priming task in which, on each trial, a briefly flashed and visually masked prime word was shown just before a target word. Primes were either congruent with the category of the following target or incongruent. In analyzing data, we calculated priming as the difference in accuracy between congruent and incongruent trials.
- The categorization task was speeded by use of a response window (RW) which, as in earlier work (Draine & Greenwald, 1998), obliges participants to respond quickly so as to maximize responding during the fleeting activation elicited by the masked primes.
- 18 participants were presented primes at only one duration, 33 ms, while 26 participants received 33 ms primes on half of trials and 100 ms on the other half. The shorter-duration primes are normally consciously imperceptible, while the 100 ms are visible for most participants (but this study did not directly test conscious perceptibility).
- As part of an ongoing series of studies examining subword processing in masked priming, the current research manipulated the order of letters in the words presented as primes. For the correlations reported under Results, no differences were observed between the type of primes, and the correlations collapse across prime type.
- After the priming task, participants completed a survey that assessed their COVID-19 history, including time since infection, severity of symptoms, and severity of brain fog.

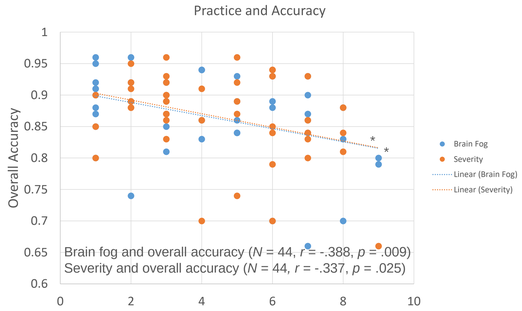
Trial sequence in priming tasks

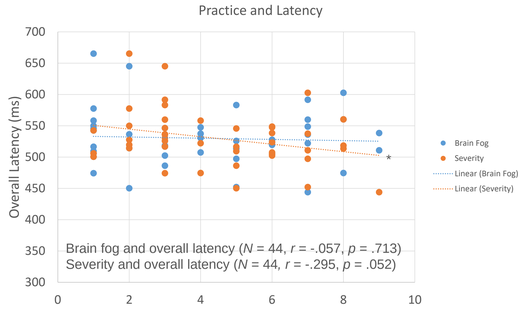


Results

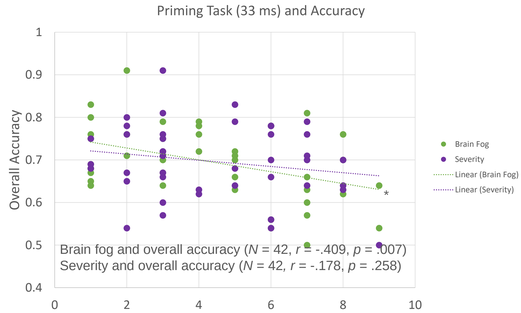
There was no significant correlation between priming magnitude and either severity (p = .545) or brain fog (p = .290). However, there was evidence of global cognitive deficits, as depicted in the scatterplots below.

Practice

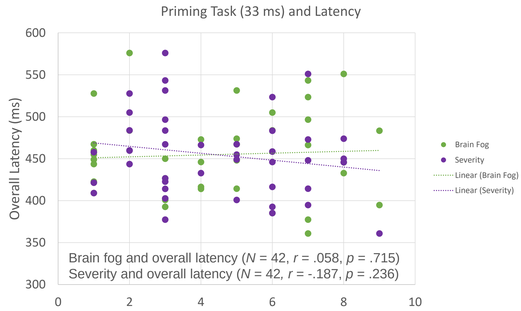




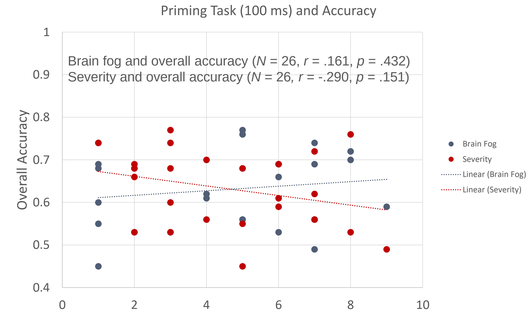
Priming Task (33ms)

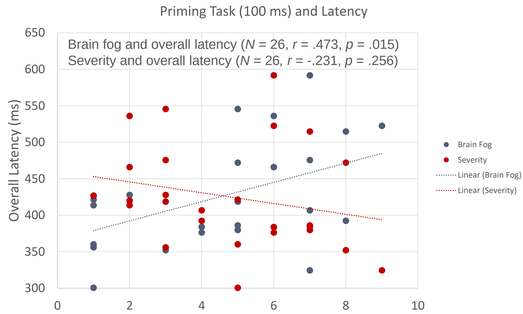


*Overall accuracy refers to the average accuracy from congruent and incongruent trials



Priming Task (100 ms)





Discussion

• Strongest evidence for impairment was observed in the practice phase, where both brain fog and severity were significantly associated with reduced accuracy *(p* = .009 and .025 respectively), and severity also showed a marginally significant association (p = .052) with faster reaction time (possibly suggesting an inability to withhold incorrect responses). The fact that evidence of impairment was strongest in the practice phase may indicate that prior-infected participants have a harder time adjusting to a novel task than participants without prior infection.

• Impairment in accuracy of speeded responding associated with brain fog was also observed in the priming task when primes were 33 ms and presumably consciously imperceptible.

• When primes were 100 ms in duration, there was no evidence for impairment of accuracy, but brain fog was significantly associated with *slower* reaction times (p = .015). A straightforward interpretation of this finding is that when the primes are visible, as they are at 100 ms, participants are more distracted by them, and prior-infected participants have more difficulty in shedding the distracting effect of the more salient 100 ms primes.

• As all but the first block of practice involved speeded responding—responding that for most participants is experienced as effortful and requiring attentive cognitive exertion—the present findings hold promise that speeded-response tasks may be sensitive measures of cognitive sequelae of prior COVID-19 infection potentially not revealed in other kinds of tasks.

References

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Tavares-Júnior, J. W. L. et al (2022). COVID-19 associated cognitive impairment: A systematic review. *Cortex*.

Acknowledgements

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